SMS Spam Detection Using Machine Learning and NLP

Alfido Tech Artificial Intelligence Internship — Task 1

Project Overview

This project develops an SMS spam detection system using Natural Language Processing (NLP) and supervised Machine Learning. The workflow covers data cleaning, exploratory analysis, TF-IDF feature extraction, model comparison, cross-validation, evaluation, and custom predictions.

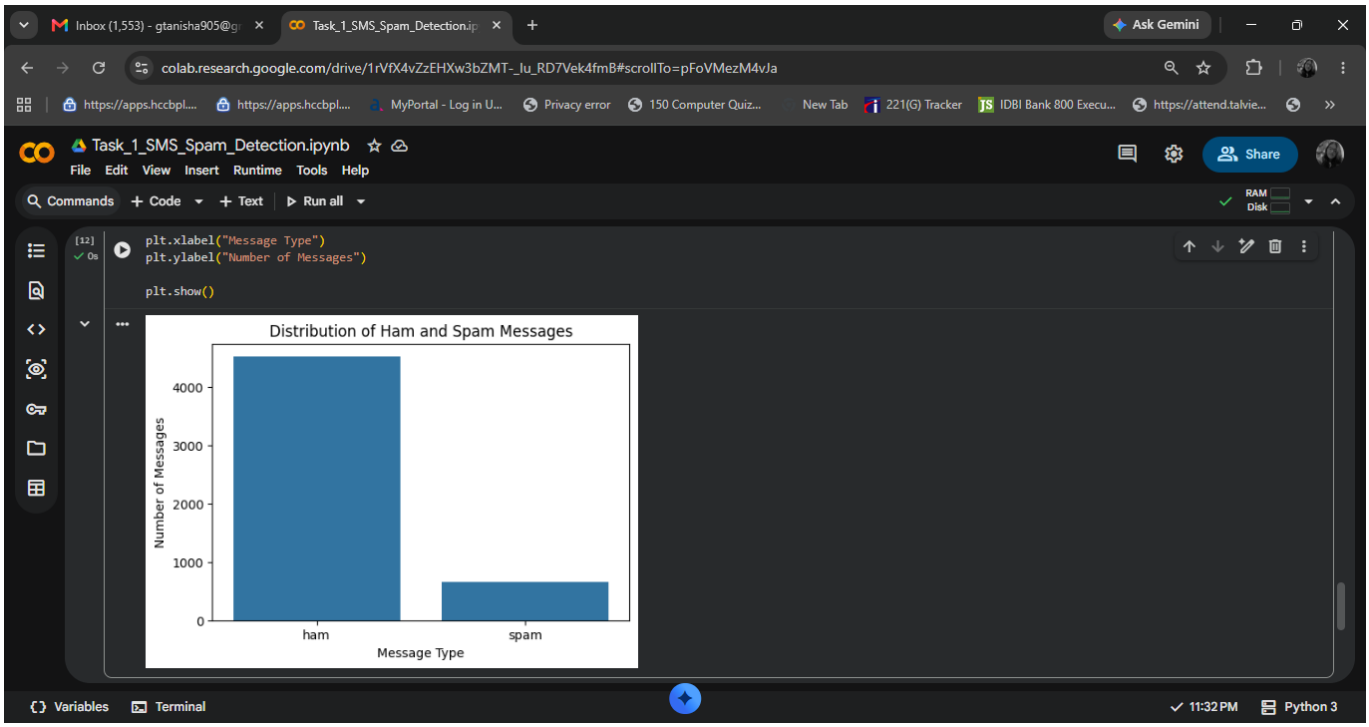
Dataset

SMS Spam Collection Dataset from the UCI Machine Learning Repository. After duplicate removal, cleaned SMS messages were used for binary classification: ham = 0 and spam = 1.

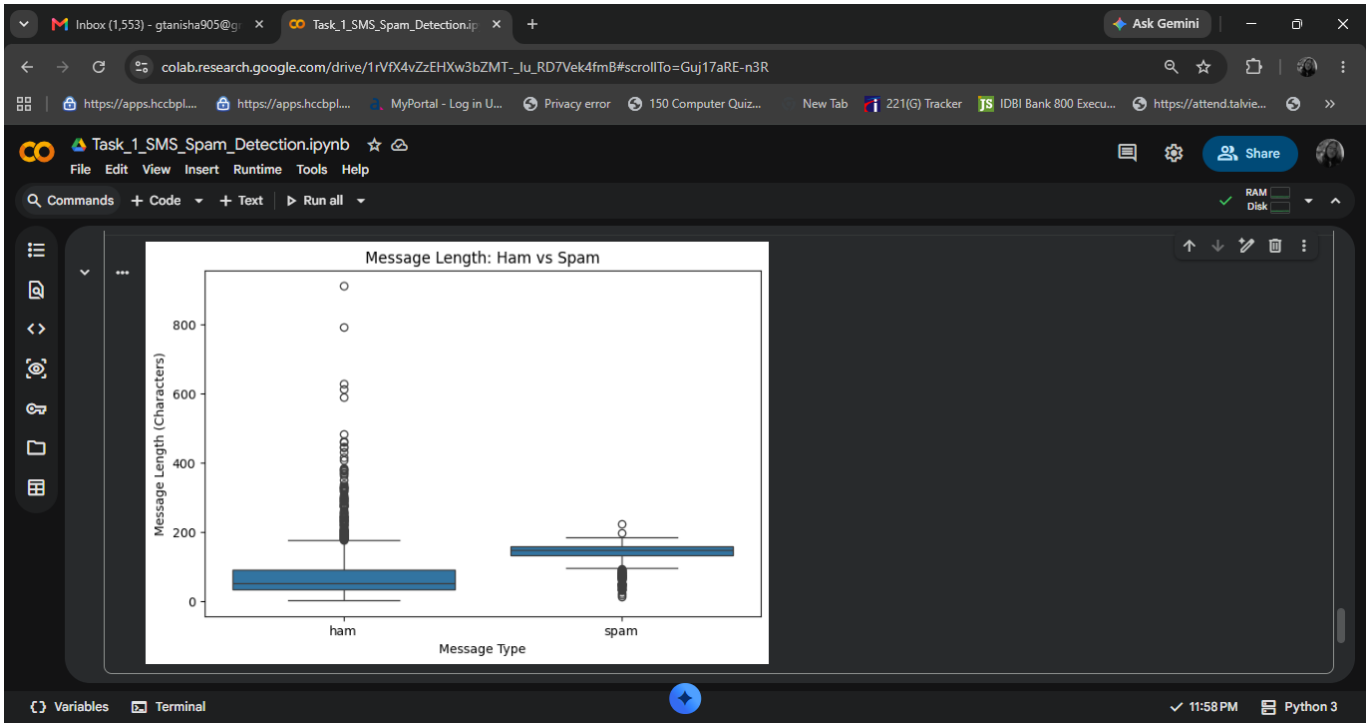
Methodology

1. Data preprocessing and duplicate removal
2. Exploratory Data Analysis (EDA)
3. Stratified 80/20 train-test split
4. TF-IDF vectorization using unigrams and bigrams
5. Logistic Regression and Random Forest classification
6. 5-fold Stratified Cross-Validation
7. Evaluation using Accuracy, Precision, Recall, F1-Score and ROC-AUC

Exploratory Data Analysis

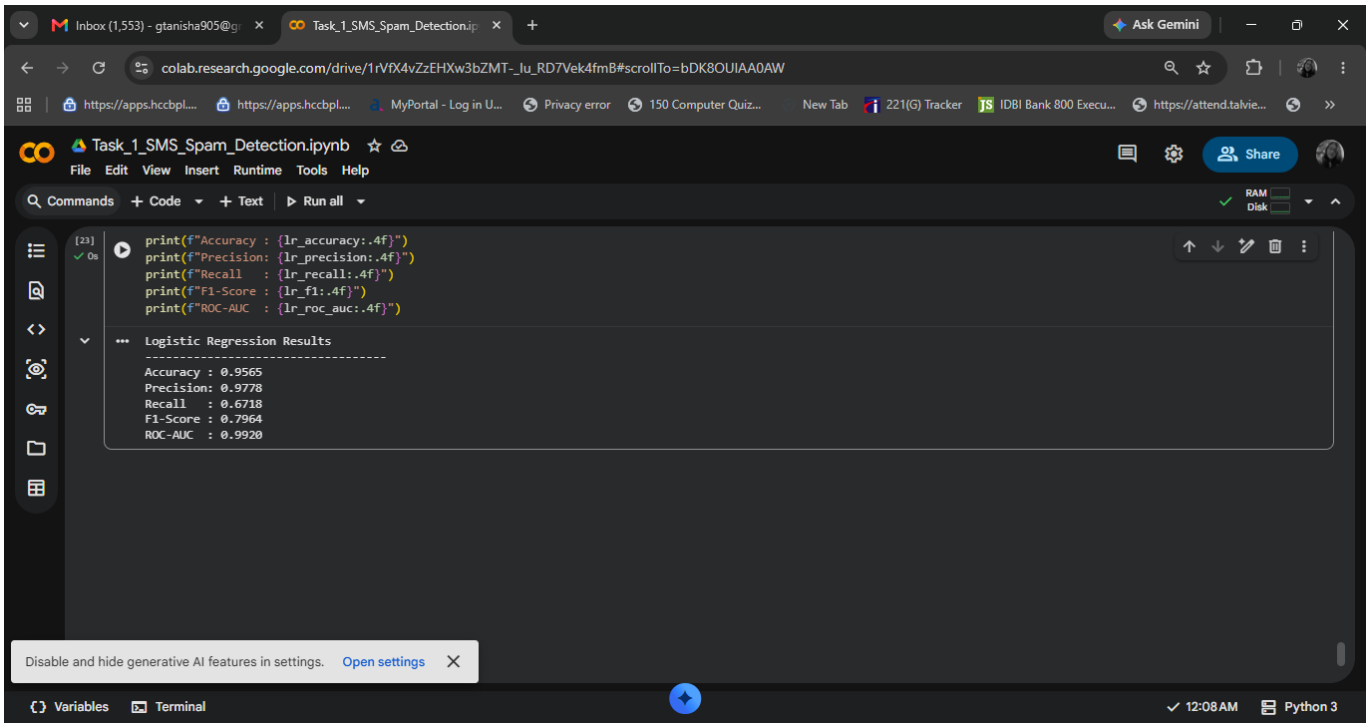


Class Distribution

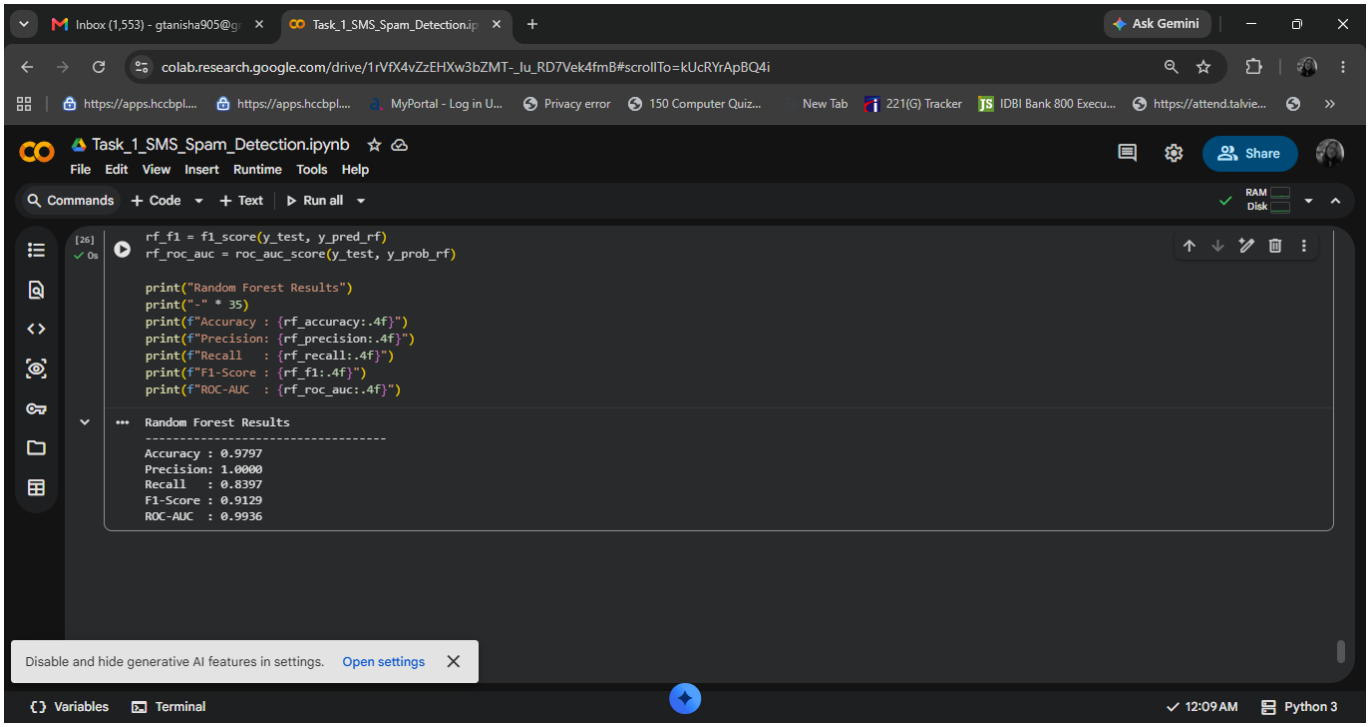


Message Length Analysis

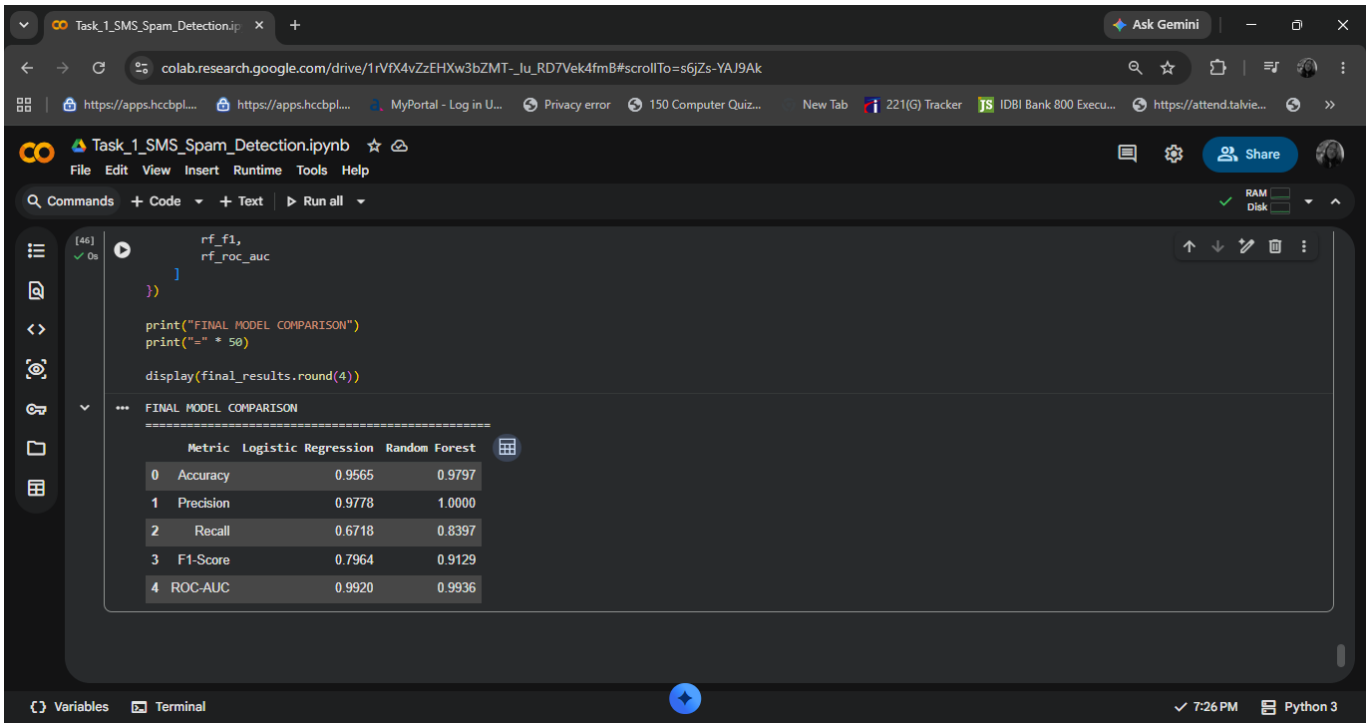
Model Training and Evaluation



Logistic Regression Results



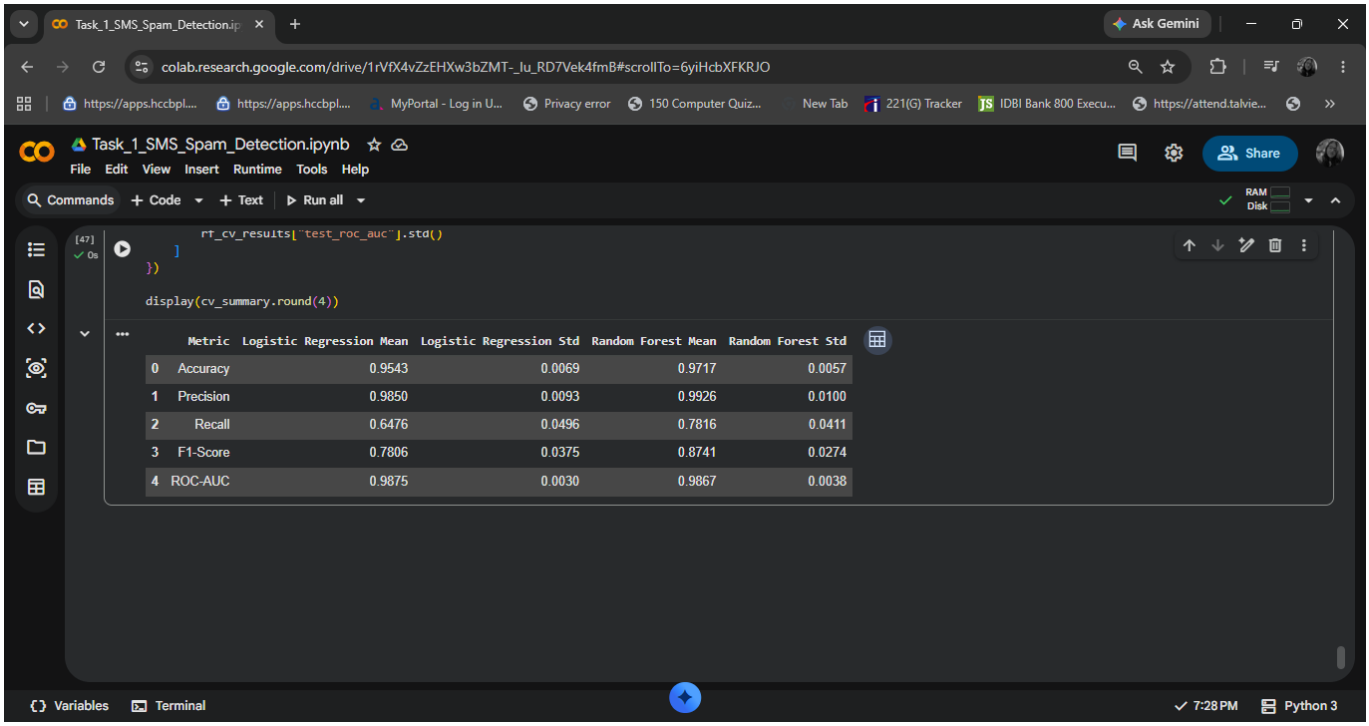
Random Forest Results



Model Comparison

Cross-Validation Results

Five-fold stratified cross-validation was used to assess the stability of both models. Random Forest showed stronger overall cross-validation accuracy, precision, recall and F1-score, while Logistic Regression had a slightly higher mean ROC-AUC.

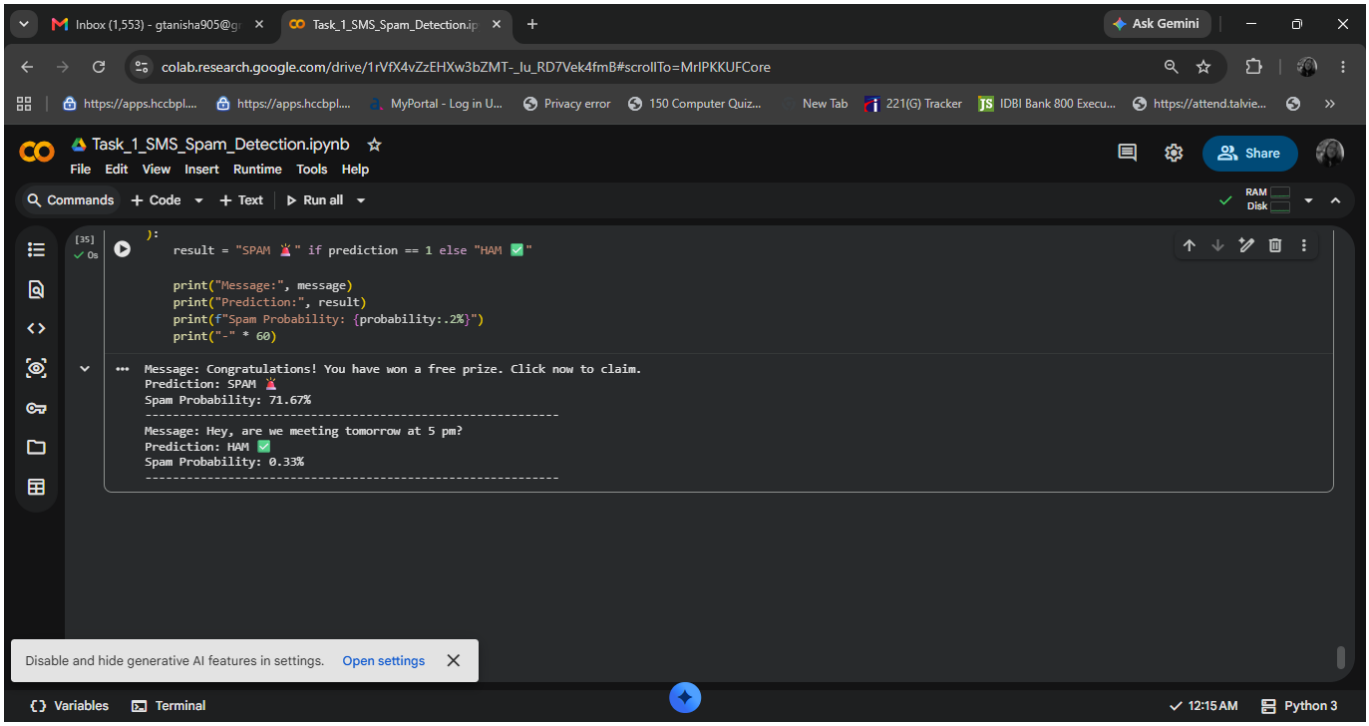


Metric	Logistic Regression Mean	Logistic Regression Std	Random Forest Mean	Random Forest Std
0 Accuracy	0.9543	0.0069	0.9717	0.0057
1 Precision	0.9850	0.0093	0.9926	0.0100
2 Recall	0.6476	0.0496	0.7816	0.0411
3 F1-Score	0.7806	0.0375	0.8741	0.0274
4 ROC-AUC	0.9875	0.0030	0.9867	0.0038

Cross-Validation Summary

Final Model and Custom Predictions

Random Forest was selected as the final model based on its overall test performance and cross-validation results.



Custom Predictions

Final Random Forest Performance

Metric	Score
Accuracy	97.97%
Precision	100.00%
Recall	83.97%
F1-Score	91.29%
ROC-AUC	99.36%

Conclusion

The project successfully demonstrates an end-to-end SMS spam classification workflow using NLP and Machine Learning. Random Forest achieved 97.97% accuracy, 100% precision, 83.97% recall, 91.29% F1-score and 99.36% ROC-AUC on the test set. The trained model and TF-IDF vectorizer were saved for future inference and deployment.

Project Files

Task_1_SMS_Spam_Detection.ipynb
spam_random_forest_model.pkl
tfidf_vectorizer.pkl
README.md